

NVIDIA HGX AI Factory — Enterprise Reference Architecture

A consolidated document compiled from the NVIDIA Enterprise Reference Architecture documentation.

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1. Abstract

This NVIDIA Enterprise Reference Architecture (RA) is a practical design guide for an NVIDIA HGX AI Factory. It is based on a 2-8-9-800 infrastructure configuration (2 CPUs, 8 GPUs, 9 NICs at 800 Gb/s bandwidth per GPU) with NVIDIA HGX™ B300 Servers featuring eight NVIDIA Blackwell Ultra GPUs connected via fifth-generation NVLink with 14.4 TB/s total interconnect bandwidth, along with NVIDIA ConnectX-8 SuperNICs, NVIDIA BlueField®-3 DPUs, and NVIDIA Spectrum-X™ Ethernet networking at the 32, 64, and 128 node design points.

The NVIDIA HGX AI Factory is designed to support enterprise AI inference workloads with industry-leading performance in an air-cooled form factor, AI training and fine-tuning, and high-performance computing (HPC) applications.

This document outlines the hardware components that define this scalable and modular architecture. This includes guidance regarding the SU design and specifics of Ethernet fabric topologies.

2. Overview

The NVIDIA Enterprise RA using the 2-8-9-800 node architecture with NVIDIA HGX B300 and NVIDIA Spectrum-X Networking is optimized for workloads of multi-node AI or hybrid applications. The Enterprise RA is a modular architecture based on NVIDIA-Certified HGX B300 systems, each with eight B300 SXM GPUs. Using a four-node scalable unit (SU) — see Figure 8 — this Enterprise RA scales up to 128 NVIDIA-Certified HGX B300 systems for a total of 1024 B300 SXM GPUs.

- A fully tested system scales to 32 SUs (Scalable Units). Larger clusters can be built based on customer requirements.
- Flexible rail-optimized end-of-row network architecture that can accommodate modifications in the rack layout and number of servers per rack.

Hardware support is available through the fulfillment system and channel partners. Software support from NVIDIA is based on a per-GPU paid subscription of NVIDIA AI Enterprise and NVIDIA Run:ai.

The NVIDIA HGX B300 and NVIDIA Spectrum-X Networking Platform Enterprise RA enables the following use cases:

- **AI Inference** — Large (per node) and Medium (per GPU) model parameter inference workload.
- **AI Training** — Large to small model training and fine tuning based on cluster sizing.
- **Large-scale data analytics** — GPU-accelerated big data analytics with simplified ETL processes and fewer data transfer bottlenecks.

Sizing guides provide guidance on typical characteristics and models supported for various workloads.

For all use cases, this architecture is ideal for multi-user, single-tenant workloads. Specifically, the logical design and software is streamlined for deployment and maintenance ease by tailoring the configuration to one where users are all part of the same enterprise, and accounting and access control can be consolidated.

Similarly, Kubernetes is the modern foundation of enterprise AI work, and this Enterprise RA is architected for deploying Kubernetes and Kubernetes-dependent applications and tooling.

3. Components

This section describes the hardware components of the Enterprise RA.

NVIDIA HGX B300 Baseboard

The HGX B300 baseboard (Figure 1) is an AI powerhouse that enables enterprises to expand the frontiers of business innovation and optimization. HGX B300 is a unified platform consisting of 8 Blackwell GPUs internally connected by NVLink and externally connected by a network interface of 800 Gb/s (2 x 400 Gb/s Ethernet) per GPU. This connectivity is provided by 8x ConnectX-8 SuperNICs on the HGX baseboard for the East/West networking.

Figure 1: HGX B300 Baseboard.



Table 1: HGX H200, B200 and B300 GPU per Node & Memory specifications.

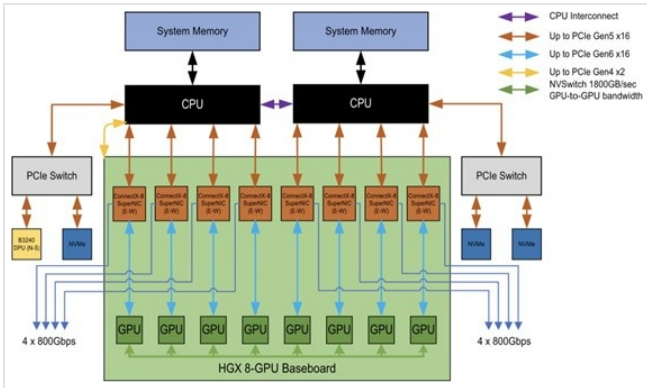
Specification	NVIDIA H200 SXM	NVIDIA B200 SXM	NVIDIA B300 SXM
Memory per GPU	141 GB HBM3e	180 GB HBM3e	288 GB HBM3e
Memory per Node	1.1 TB HBM3e	1.44 TB HBM3e	2.30 TB HBM3e
GPU Bandwidth	4.80 TB/s	Up to 8 TB/s	Up to 8 TB/s
GPU Aggregate Bandwidth per Node	38.4 TB/s	Up to 64 TB/s	Up to 64 TB/s

The HGX B300 baseboard provides up to 144 petaflops of processing power, making it a leading accelerated platform for AI. It offers advanced networking up to 800 Gbps with ConnectX-8 SuperNICs, enabling AI cloud networking, composable storage, zero-trust security, and GPU compute elasticity for AI workloads.

NVIDIA HGX B300 Systems

NVIDIA-Certified HGX B300 systems are based on a common system design with flexibility for optimizing the configuration to match the cluster requirements. An example of this design is shown in Figure 2.

Figure 2: HGX B300 System.



Elements for an NVIDIA-Certified HGX B300 system are listed in Table 2.

Table 2. NVIDIA HGX B300 system elements.

Parameter	Requirement
Target workloads	Large Language Model; Traditional DL Inference Models, HPC
GPU configuration	Eight NVIDIA B300 GPUs on an HGX B300 baseboard with up to 2304 GB of GPU memory
NVIDIA® NVLink™ and NVSwitch™	NVIDIA HGX B300 Baseboards use a combination of fifth-generation NVSwitch and fifth-generation NVLink. Total Aggregate Bandwidth 14.4 TB/s. GPU-to-GPU Bandwidth 1800 GB/s
CPU sockets	2
CPU speed	2.0 GHz minimum base CPU clock
CPU cores	Minimum of 48 physical CPU cores per socket. Recommendation of 56 physical CPU cores per socket
System memory (total across all CPU sockets)	Minimum of 2 TB system memory; minimum of 500 GB/s memory bandwidth. For optimal performance, system memory should be evenly distributed across all CPU sockets and memory channels and should be fully populated and symmetrically placed on all CPU memory controller (MC) channels.
DPU (North/South)	One NVIDIA® BlueField®-3 DPU per server

PCI Express	Eight Gen5 x16 links and one Gen4 x2 link per NVIDIA HGX B300 baseboard. One Gen5 x16 link per DPU, SuperNIC or adapter
PCIe topology	Balanced PCIe topology with connectivity spread evenly across CPU sockets and PCIe root ports
Network Adapters/NICs speed (East/West)	Eight NVIDIA® ConnectX-8 SuperNICs per NVIDIA HGX B300 baseboard. Up to 800 Gbps per adapter
Local storage	Inference Servers: Minimum 1 TB NVMe drive per CPU socket. Training / DL Servers: Minimum 2 TB NVMe drive per CPU socket. HPC Servers: Minimum 1 TB NVMe drive per CPU socket. 1 TB NVMe boot drive
Remote systems management	SMBPBI over SMBus (OOB) protocol to BMC; PLDM T5-enabled; SPDMM-enabled
Security	TPM 2.0 module (secure boot)

Control Plane / Management Nodes

The cluster design specified in this Enterprise RA can support up to eight control plane nodes. In addition to the standard compute nodes, control plane nodes are needed to run the software that manages the cluster and provides access for users. The specific components depend on the management software. Table 3 shows an example configuration for these nodes.

For example, a configuration using NVIDIA Base Command Manager, Slurm, and Kubernetes together can include seven control plane nodes in total: two for Base Command Manager (with high availability configured), two for Slurm head nodes, and three for Kubernetes control plane nodes. This uses seven of the eight available control plane nodes possible.

Table 3. Control plane node components

Component	Quantity	Description
CPU	2	32C Intel Xeon Gold 6448Y or equivalent; 32C AMD EPYC 9354
North/South (DPU)	1	NVIDIA BlueField-3 B3220 DPU with two 200G ports and 1 Gb RJ45 management port. Other variants can be supported as per Compute Node alternatives — Table 4
System Memory	—	Minimum of 256 GB DDR5
Boot Drive	1	1 TB NVMe SSD
Local storage	1	4 TB NVMe SSD. More may be required if image storage is required
BMC	1	1 Gb RJ45 management port

In cases where the existing control plane nodes are missing, the Enterprise RA recommends deploying one set per cluster. Configure these nodes for high availability to keep your cluster running smoothly.

Networking

The NVIDIA HGX platform networking configuration enables the highest AI performance and scale, while ensuring cloud manageability and security. It leverages NVIDIA expertise in AI cloud data centers and optimizes network traffic flow:

- **East/West (Compute Network) traffic:** This refers to traffic between NVIDIA HGX systems within the cluster, typically for multi-node AI training, AI fine tuning, HPC collective operations, and other workloads.
- **North/South (Customer and Storage Network) traffic:** This involves traffic between NVIDIA HGX systems and any external resources including cloud management and orchestration systems, remote data storage nodes, and other parts of the data center or the Internet.

When combined with NVIDIA Spectrum-X Ethernet, the NVIDIA HGX B300 platform delivers strong performance for deep learning training and inference, data science, scientific simulation, and other modern workloads. The following sections describe recommended NVIDIA HGX B300 platform configurations along with their associated NVIDIA networking platforms.

Compute (Node East/West) Ethernet Networking

On the NVIDIA HGX B300 baseboard, NVIDIA ConnectX-8 SuperNIC offers up to 800 Gb/s, low-latency network connectivity between GPUs in the AI cluster, featuring RDMA and RoCE acceleration, with NVIDIA® GPUDirect® and GPUDirect Storage technologies. For data centers that deploy Ethernet, NVIDIA ConnectX-8 offers a range of advancements including RoCE optimizations and multi-tenancy, which are key in AI cloud environments. Additionally, NVIDIA ConnectX-8 offers advanced hardware offloads for overlay networks such as VXLAN and NVGRE, which help reduce CPU load and improve network efficiency.

Beyond raw speed, NVIDIA ConnectX-8 enhances network performance with features like advanced congestion control, telemetry-based routing, and quality of service (QoS) capabilities to meet diverse AI application and training requirements. Security is built in with hardware acceleration for cryptographic protocols like IPsec and MACSec, taking these tasks over from the CPU to maintain performance.

Multi-node deployments with an NVIDIA HGX B300 platform should adhere to the following total compute network bandwidth per GPU recommendations.

The NVIDIA ConnectX-8 SuperNIC is integrated onto the NVIDIA HGX B300 baseboard, maintaining a 1:1 GPU-to-NIC ratio to ensure optimal bandwidth and direct connectivity for each GPU.

Total Minimum Compute Network Bandwidth

- 400 GB/s (8x 400 Gb/s Ethernet NICs)

Total Recommended Compute Network Bandwidth

- 800 GB/s (16x 400 Gb/s Ethernet NICs using breakout)

Converged (Node North/South) Ethernet Networking

This section describes the NVIDIA BlueField-3 role for the North/South Ethernet network in the NVIDIA HGX platform and the recommended NVIDIA BlueField-3

models for this infrastructure.

The NVIDIA BlueField-3 data processing unit (DPU) is a 400 Gb/s infrastructure compute platform that enables organizations to securely deploy and operate NVIDIA HGX AI data centers at massive scales. NVIDIA BlueField-3 DPU is optimized for the North/South network and the NVIDIA BlueField-3 IC is optimized for the East/West AI compute fabric.

NVIDIA BlueField-3 offers several essential capabilities and benefits within the NVIDIA HGX platform:

- **Workload Orchestration:** NVIDIA BlueField-3 serves as an optimized compute platform for the data center control-plane, enabling automated provisioning and elasticity. This empowers NVIDIA HGX AI cloud platforms to scale resources dynamically based on fluctuating demand, ensuring efficient allocation of computing resources for transient AI workloads.
- **Storage Acceleration:** NVIDIA BlueField-3 provides advanced storage acceleration features that optimize data storage access. Its innovative NVIDIA BlueField SNAP technology enables remote storage devices to function as local, improving AI performance and streamlining cloud operations.
- **Secure Infrastructure:** NVIDIA BlueField-3 operates in a highly secure zero-trust mode and functions independently from the host, significantly enhancing the security of the NVIDIA HGX platform. In addition, NVIDIA BlueField-3 enables a wide range of accelerated security services, including next-generation firewall and micro-segmentation, bolstering the overall security posture of the infrastructure.

The NVIDIA BlueField-3 DPU integration enhances NVIDIA HGX platforms by improving resource management, performance, security, and scalability, making them well-suited for AI workloads in datacenter environments.

NVIDIA BlueField-3 DPUs support multiple operating modes, including Embedded Function (ECPF) or DPU mode, which is commonly used as a default. In this mode, the Arm subsystem on the DPU owns and manages NIC resources, and network traffic typically passes through a virtual switch on the DPU before reaching the host, adding an extra layer of control.

To support secure management of NVIDIA HGX platforms, many NVIDIA BlueField products integrate an onboard Baseboard Management Controller (BMC). This BMC enables provisioning and management of both the NVIDIA BlueField DPU and the HGX platform using standard tools such as Redfish APIs, relies on an external root-of-trust for firmware protection, and connects to the management network through a dedicated 1 GbE out-of-band port.

When planning NVIDIA HGX deployments with NVIDIA BlueField-3, it is important to account for card form factor and power requirements. Some BlueField-3 configurations for north-south connectivity can draw more than 75 W and therefore require both standard PCIe slot power and an additional PCIe power connector rated for at least 75 W.

For NVIDIA BlueField-3 DPUs used in the North/South network for NVIDIA HGX 8-GPU platforms, the NVIDIA BlueField-3 products listed in Table 4 are generally recommended as suitable options.

Table 4. HGX B300 recommended BF-3 DPUs for North/South Network

Product	PCIe Card Form Factor	Connector Required	Applicable Topologies
NVIDIA BlueField-3 B3240 P-Series FHHL DPU, 400GbE (default mode) / NDR IB, Dual-port QSFP112	Single-slot FHHL PCIe Gen5 x16 with x16 PCIe extension option	8-pin ATX 12V PCIe; 1G/USB Management Interfaces; QSFP112 connector cages	Cold-aisle slots/servers only
NVIDIA BlueField-3 B3240 P-Series FHHL DPU, 400GbE (default mode) / NDR IB, Dual-port QSFP112	Dual-slot FHHL PCIe Gen5 x16 with x16 PCIe extension option	8-pin ATX 12V PCIe; 1G/USB Management Interfaces; QSFP112 connector cages	All
NVIDIA BlueField-3 B3220 P-series FHHL DPU, 200GbE (default mode) / NDR200 IB, Dual-port QSFP112	Single-slot FHHL PCIe Gen5 x16 with x16 PCIe extension option	8-pin ATX 12V PCIe; 1G/USB Management Interfaces; QSFP112 connector cages	All

Note: NVIDIA recommends using the NVIDIA BlueField-3 B3240 DPU with NVIDIA HGX B300 platforms rather than the NVIDIA BlueField-3 B3220 that is common in NVIDIA HGX H100/200 and HGX B200 systems with 2x 200 GbE uplinks. For HGX B300 deployments, it can be helpful to consider future workloads such as distributed inference that may use KV cache offloads to high-speed, network attached storage. In these cases, having higher burst I/O capacity per GPU can be advantageous, even if the average bandwidth needs for typical training workloads are not very high.

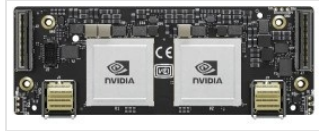
4. Networking Hardware

This section describes the networking hardware used to connect elements of the Enterprise RA.

NVIDIA ConnectX-8 HCA (on baseboard)

With support for Ethernet networking up to 800 Gb/s (2x 400 Gb/s), the NVIDIA ConnectX-8 SuperNIC (Figure 3) delivers extremely fast, efficient network connectivity, and significantly enhances system performance for AI factories and cloud data center environments. Depending on the deployment scenario and use case, the SuperNIC can be leveraged for dual-plane east-west networking (using 2x 400 Gb/s twin-port transceivers) or single-plane east-west networking (using 1x 400 Gb/s single-port OSFP transceivers) enabling flexible support for different performance and bandwidth requirements.

Figure 3. NVIDIA ConnectX-8 HCA.



NVIDIA BlueField-3 B3240 DPU

The NVIDIA BlueField-3 B3240 DPU with two 400 GbE ports (Figure 4) is used in the in-band management and storage fabric. The Enterprise RA recommends the use of dual-ports for best performance and flexibility.

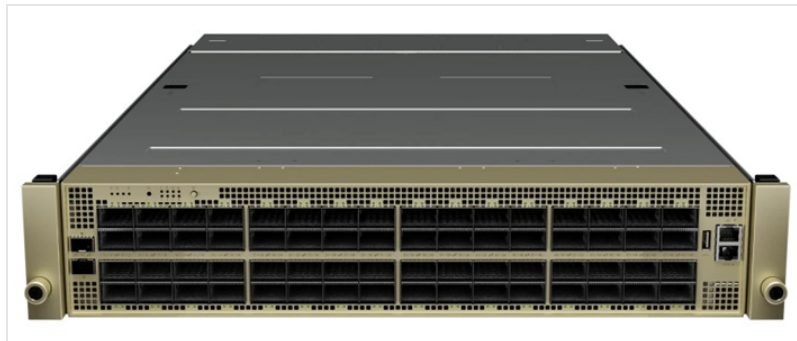
Figure 4. NVIDIA BlueField-3 B3240 DPU



NVIDIA SN5610 Switch

The NVIDIA SN5610 switch (Figure 5) offers 64 total ports of 800 Gbps to provide connectivity for Compute (E/W), Customer and Storage (N/S), in-band management.

Figure 5. NVIDIA SN5610 Ethernet switch.



NVIDIA SN2201 Switch

The NVIDIA SN2201 switch (Figure 6) offers 48 ports to provide connectivity for out-of-band (OOB) management. OOB management provides consolidated management connectivity for all components in the Enterprise RA.

Figure 6. NVIDIA SN2201 Ethernet switch.



5. Networking Physical Topologies

The Enterprise RA configurations use three physical network fabrics:

- Compute (Node East/West) Network
- CPU Converged (Node North/South) Network
- Out-of-Band Management Network

Each network is discussed in this section.

Compute (Node East/West) Network

The compute fabric (East/West) is built using switches with NVIDIA Spectrum technology in a full non-blocking fat-tree topology to provide the highest level of performance for the application running over the NVIDIA HGX B300 cluster. The compute fabric is an RDMA-based fabric. It is designed to provide the shortest hops through the network for the application, in a leaf-spine manner, where the GPUs are connected using a rail-optimized network topology through their respective NVIDIA ConnectX-8 SuperNICs. This design allows the most efficient communication for multi-GPU applications within and across the nodes.

The leaf and spine architecture allows for a scalable and reliable network that can fit varied sizes of clusters using the same architecture.

The compute fabric is designed to facilitate redundancy, maximize bandwidth and minimize network latency required to connect GPUs within a server and within a rail. While the architecture below recommends a dual-plane compute fabric for resiliency purposes, a cost-effective single-plane configuration is also supported based on deployment needs.

The compute network is not necessarily required for inference workloads. Most common pre-trained models do not exceed the size of a single NVIDIA B300 SXM GPU. Models beyond 120B parameters may require model parallelism and require more than one NVIDIA B300 SXM GPU. This parallelism will still reside within the same node as each server node can hold up to 8x NVIDIA B300 SXM GPUs.

For larger network design points, due to the high number of endpoints required, the Compute Fabric utilizes a spine and leaf networking architecture. This is principally to support the required design points but also to allow extensibility in the architecture.

In the architectures below, the Compute Fabric is merged with the Converged Fabric for smaller design points. This is to lower cost and to simplify the network architecture.

Compute Fabric Excluded for Pure Inference Deployments

For the use case of pure inference, a compute network may not be necessary. Each NVIDIA B300 SXM GPU can support a maximum model size of approximately 120B parameters. Also, tensor parallelism yields less performant results per GPU than running the model on the same GPUs in data parallel mode.

Since there are no performance gains from running an inference model on multiple GPUs, either on a single node or across multiple nodes, a compute network is not necessary.

The drawback to implementing an infrastructure without a compute network is that the infrastructure cannot be used for any hybrid workflows including model training. The infrastructure can be deployed without a compute network and can be retrofitted later, albeit with potentially significant downtime and reconfiguration.

Multi-Plane Topology Approach

While each GPU is associated with a dedicated NVIDIA ConnectX-8 SuperNIC, the coupling of a single link from a NIC to a single point of failure (SPOF) in the fabric is a concern. Increasingly, customers are opting for a load-balanced multi-link approach which can satisfy both the full capacity of the GPU bandwidth as well as avoiding a SPOF. Each compute networking plane forms a separate fabric, where the resiliency and the load balancing between the two planes is handled by the NCCL on the host. Any failure to provide GPU-to-GPU connectivity via one plane will seclude traffic to the alternative plane, but if both planes are active, the traffic will be balanced between planes in a manner that utilizes the bandwidth of both links. Load balancing via NCCL is more efficient than other L2/L3 bonds like LAG that only have "local" awareness for traffic balancing and static hash-based distribution which is suboptimal for AI workloads due to low entropy.

A second benefit of plane separation is expansion of the fan-out of the NIC and switches. A higher radix of interfaces enables the creation of a larger network with fewer network tiers in a Clos topology.

Dual Plane Topology

In this Enterprise RA we recommend using a Dual Plane topology. A Dual Plane topology involves building two identical planes that connect to each GPU interface. With each GPU generating 800 Gb/s bandwidth through the NVIDIA ConnectX-8 SuperNICs, dual plane topology involves breaking the interface to 2x 400 Gb/s interfaces. Every such interface is then connected to a different leaf switch, and every such leaf switch is part of an independent fabric that scales to 1024 interfaces of 400 Gb/s as part of this reference architecture. The GPU is not aware of the two independent fabrics, each carrying 50% of the generated traffic, and is only aware that 100% of traffic is carried by the NVIDIA ConnectX-8 SuperNIC. Tracking of each plane, load balancing, and failure handling is handled by the ConnectX-8 SuperNIC on the hardware level. A failing or degraded plane will carry an impact linearly associated with the dropped bandwidth.

Single Plane Topology

A Single Plane topology is also possible as a cost-effective alternative to the Dual Plane topology. In the configuration, each GPU operates at 400 Gb/s of bandwidth through the NVIDIA ConnectX-8 SuperNIC without breaking the interface — using a single 1x 400 Gb/s connection instead of the 2x 400 Gb/s. Each interface is connected to a leaf switch within a single compute fabric that scales to 1024 interfaces of 400 Gb/s as part of the reference architecture. While this approach reduces the total GPU bandwidth by 50%, it is well-suited for workloads that do not warrant maximum throughput and can benefit from lower networking infrastructure costs. The use of OSFP transceiver modules enables seamless migration between single plane and dual plane topologies, allowing flexibility in adapting to evolving performance needs.

CPU Converged (Node North/South) Network

A converged network for both storage and in-band management are used in the Enterprise RA. This provides enterprises with flexible storage allocation and easy network management. The converged network has the following attributes:

- Provides high bandwidth to shared storage and connects the customer through the converged network.
- Is independent of the compute fabric to maximize both storage and application performance.
- Each compute and management node is connected with two 400 GbE ports to two separate switches to provide redundancy and high storage throughput that can reach up to 40 GB/s per node.

The fabric is built on Ethernet technology with RDMA over Converged Ethernet (RoCE) support and utilizes NVIDIA BlueField-3 B3240 DPUs in each compute node to deliver existing and emerging cloud and storage services.

- It is flexible and can scale to meet specific capacity and bandwidth requirements.
- Tenant-controlled management nodes provide the tenant the flexibility to deploy OS and their job scheduler of choice.
- Hybrid storage fabric design with support for tenant isolation provides access to both shared and dedicated storage per tenant.
- Used for node provisioning, data movement, internet access, and other services that must be accessible by the users.

Out-of-Band Management (Node) Network

The OOB management network connects all the Baseboard Management Controller (BMC) ports, as well as other devices that should be physically isolated from system users to allow infrastructure management. This includes the 1 GbE switch management ports and the NVIDIA BlueField-3 DPU management ports.

6. Networking Logical Architecture

This Enterprise RA uses a converged spine-leaf network providing physical fabrics for the use cases such as inferencing and fine-tuning. Deeper explanations of each of the fabric roles are below:

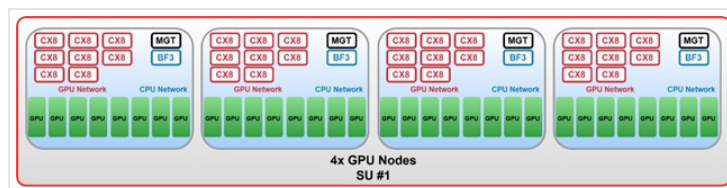
- **GPU Compute (East/West) Network** — The GPU Compute (East/West) network is an RDMA-based fabric in a leaf-spine architecture where the GPUs are connected using a rail-optimized network topology through their respective SuperNICs. This design allows the most efficient communication for multi-GPU applications within and across the compute nodes due to the high number of endpoints needed. The leaf-spine networking architecture also allows for future extensibility.
- **CPU Converged (North/South) Network** — The CPU Converged (North/South) network connects the nodes using two 400 GbE ports to two separate switches to provide redundancy and high storage throughput. This network is used for node communications with compute, storage, in-band management, and end-user connections.
- **Storage (North/South) Connectivity** — This network specifically provides converged connectivity for storage infrastructure. Principally storage is attached on this network and then can be provided to the nodes via the CPU Compute network as required.
- **Customer (North/South) Network Connectivity** — Typically, this is upstream connectivity to connect the cluster to the rest of an enterprise customer's networking infrastructure. This is provisioned to provide ample bandwidth for typical user communications.
- **Support Server Networking** — This is dedicated networking for the respective support servers. These servers generally provide management, provisioning, monitoring, and control services to the rest of the cluster. High-performance networking is used here due to requirements such as cluster deployment and imaging.
- **Out-of-band Management Networking for the infrastructure** — All infrastructure requires management. This network provides bulk management 1Gb RJ45 connectivity for all the nodes. This network uses low-cost bulk management switches. These switches have upstream connections to the core networking infrastructure to allow flexibility and wider consolidation of services such as management and monitoring.

Note: VLAN isolation would be used to provide logical separation of the networks above over the single physical fabric.

Enterprise RA Scalable Unit (SU)

This Enterprise RA is built on scalable units (SU) based on 4 compute nodes. Each SU is a discrete entity of computation that is tied to the port availability size of the network devices. SUs can be replicated to adjust the scale of the deployment with more ease.

Figure 7: Example diagram of 4-Node SU.



The 4-Node scalable unit provides the following connectivity building blocks:

- For the Compute (East/West) fabric: 4 servers, each with 8x NVIDIA ConnectX-8 SuperNICs, providing 64x 400 Gb/s connections and a total aggregate bandwidth of 25.6 Tb/s.
- For the Converged (North/South) fabric: 4 servers, each with 1x NVIDIA BlueField-3 B3240 DPU providing 8x 400 Gb/s connections and a total aggregate bandwidth of 3.2 Tb/s.
- For the Out-of-band Management fabric, 4 servers, each with 6x 1 Gb/s connections providing 24x 1 Gb/s for management.

Spine-Leaf Networking

The network fabrics are built using switches with NVIDIA Spectrum-X Ethernet technology in a full non-blocking fat-tree topology to provide the highest level of performance for the application running over the Enterprise RA configuration. The networks are RDMA-compliant fabrics in a leaf-spine architecture where the GPUs are connected using a rail-optimized network topology through their respective SuperNICs. This design allows the most efficient communication for multi-GPU applications within and across the nodes.

The leaf and spine architecture supports a scalable and reliable network that can fit varied sizes of clusters using the same architecture. The compute network is designed to maximize bandwidth and minimize network latency required to connect GPUs within a server and within a rail.

In addition to the supported compute fabrics, the converged spine-leaf network also supports the following attributes:

- Provides high bandwidth to high performance storage and connects to the data center network.
- Each compute node is connected with two 400 GbE ports and each management node is connected with two 200 GbE ports to two separate switches to provide redundancy and high storage throughput.

Out-of-Band (OOB) Management Network

The OOB management network connects the following components:

- Baseboard Management Controller (BMC) ports of the server nodes
- BMC ports of the BlueField-3 DPU and SuperNICs
- OOB management ports of switches

This OOB network can also connect other devices that should have management connectivity physically isolated for security purposes. The NVIDIA SN2201 switch is used to connect to the BMC/OOB 1 Gbps ports of these components. Scale out this network using a 25 Gbps or 100 Gbps spine layer to connect the NVIDIA SN2201 uplink switches.

Optimized Use Case

This architecture is optimized for both inference and training use cases. Deployment of this single architecture will enable both types of workloads.

Architecture Details

The components of the NVIDIA HGX B300 Spectrum Enterprise RA using Spectrum-X 800 Gbps for the compute fabric are described in Table 5.

Table 5: HGX B300 Server Ethernet components with Spectrum-X fabric

Component	Technology
Compute node servers (4-128)	NVIDIA HGX B300 Servers with 8 NVIDIA B300 SXM GPUs, configured with: one NVIDIA BlueField-3 B3240 dual-port 400 GbE DPU; eight NVIDIA ConnectX-8 single-port 800 GbE SuperNICs, configured for 2x 400 Gb Ethernet breakout
Compute (East/West) Spine-Leaf Fabric	NVIDIA SN5600 128-port 400 GbE switches
Converged (North/South) Spine-Leaf Fabric	NVIDIA SN5600 128-port 400 GbE switches
OoB management fabric	NVIDIA SN2201 48-port 1 Gb plus 4-port 100 GbE

Compute (Node East/West) Fabric Table

Table 6 below shows the number of cables and switches required for the dual-plane compute (Node East/West) Fabric for different SU sizes. Figures shown are aggregate amounts for both planes.

Table 6: Compute (Node East/West) and switch component count

Nodes	GPUs	SUs	Leaf Switches	Spine Switches	Uplinks per leaf to spine @ 400G	Node-to-Leaf Transceivers (Compute)	Node-to-Leaf Transceivers (Switch)	Switch-to-Switch Transceivers	Compute-to-Leaf Cables	Switch-to-Switch Cables
32	256	8	8	4	32	256	256	512	512	512
64	512	16	16	8	16	512	512	1024	1024	1024
128	1024	32	32	16	8	1024	1024	2048	2048	2048

Converged (Node North/South) Fabric Table

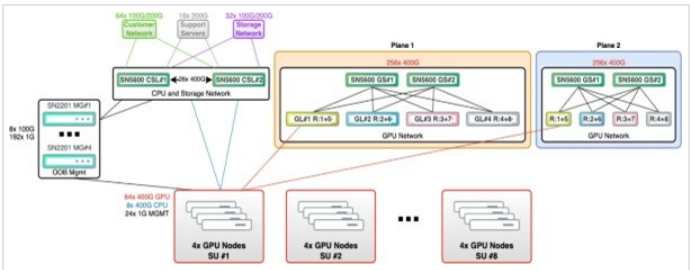
Table 7 below shows the number of cables and switches required for the Converged (Node North/South) Fabric for different SU sizes. For lower node counts such as 16 nodes where the Converged Network has been consolidated with the Compute Network, the transceivers and cables are additional to the quantities in Table 7 used for the East/West fabric.

Table 7: Converged (Node North/South) and switch component count

Nodes	GPUs	SUs	Leaf	Spine	CPU (Node N-S)	Storage	Mgmt. Uplinks	Customer	Support	ISL Ports (both ends)	Node-to-Leaf Transceivers (Node)	Node-to-Leaf Transceivers (Switch)	Switch-to-Switch Transceivers	Node-to-Leaf Cables	Switch-to-Switch Cables
32	256	8	2	N/A	32	8	4	16	4	26	188	62	26	188	26
64	512	16	4	2	64	16	4	32	4	256	352	118	256	352	256
128	1024	32	8	4	128	32	8	64	4	512	688	232	512	688	512

32 Nodes with 256 NVIDIA HGX B300 SXM GPUs

Figure 8: NVIDIA HGX B300 Spectrum Enterprise RA with 32 servers (256 GPUs) and dual-plane compute fabric.



Architecture Overview

- 32 NVIDIA HGX B300 Servers (256 GPUs) in 2-8-9-800 configuration with NVIDIA Spectrum-X compute fabric.
- 8 scalable units (SUs) with 4 server nodes per unit.

Network Design

- Cost-efficient converged two-switch fabric for CPU (North/South) Network.
- GPU Compute (East/West) Network uses separate, isolated dual-plane leaf-spine architecture for high-bandwidth, low-latency communication.
- Dual-plane rail-optimized, non-blocking Spectrum fabric with 64-port switches.
- Port breakout functionality consolidates ports while maintaining resiliency.

- 4 leaf switches per plane, each supporting 2 rails (1+5, 2+6, 3+7, 4+8).
- Each GPU has individual 400 Gb connection to each plane via 800 Gb port broken out to 2x 400 Gb.

Connectivity (Under Optimal Conditions)

- 64x 400G uplinks per SU for GPU Compute traffic (East/West).
- 8x 200G uplinks per SU for CPU traffic (North/South).
- Up to 8 support servers, dual-connected at 200 Gb.
- 64x 100G/200G connections to customer network (minimum 25 Gb bandwidth per GPU).
- 32x 100G/200G connections for storage (minimum 12.5 Gb bandwidth per GPU).

Management

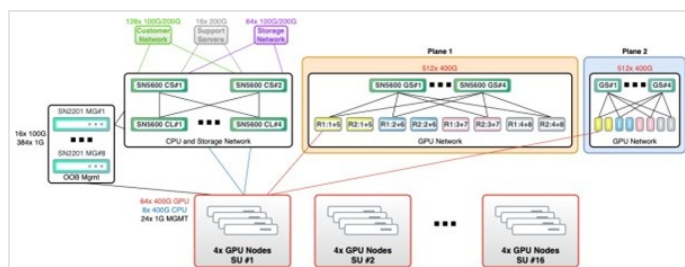
- SN2201 switch for every 2 SUs (4 switches total).
- Each SN2201 uplinked to core via 2x 100G links.
- 200 GbE management fabric with HA using eight 100 GbE core connections.
- All nodes connected to 1 Gb management switches.

Additional Considerations

- VLAN isolation separates North-South and East-West fabrics collapsed into the physical infrastructure.
- Rack layout must provide power supply redundancy; otherwise, consider an alternative rack layout.
- The number of GPU servers per rack depends on available rack power.

64 Nodes with 512 NVIDIA HGX B300 SXM GPUs

Figure 9: HGX B300 Spectrum Enterprise RA with 64 servers (512 GPUs) with dual-plane compute fabric.



Architecture Overview

- 64 NVIDIA HGX B300 Servers (512 GPUs) in 2-8-9-800 configuration with NVIDIA Spectrum-X compute fabric.
- 16 scalable units (SUs) with 4 server nodes per unit.

Network Design

- Cost-efficient converged two-switch fabric for CPU (North/South) Network.
- GPU Compute (East/West) Network uses separate, isolated dual-plane leaf-spine architecture for high-bandwidth, low-latency communication.
- Dual-plane rail-optimized, non-blocking Spectrum fabric with 64-port switches.
- Port breakout functionality consolidates ports while maintaining resiliency.
- 8 leaf switches (in 2 blocks of 4) per plane, each supporting 2 rails (1+5, 2+6, 3+7, 4+8).
- 2 leaf switches supporting each pair of rails with corresponding spine increase.
- Each GPU has individual 400 Gb connection to each plane via 800 Gb port broken out to 2x 400 Gb.

Connectivity (Under Optimal Conditions)

- 64x 400G uplinks per SU for GPU Compute traffic (East/West).
- 8x 200G uplinks per SU for CPU traffic (North/South).
- Up to 8 support servers, dual-connected at 200 Gb.
- 128x 100G/200G connections to customer network (minimum 25 Gb bandwidth per GPU).
- 64x 100G/200G connections for storage (minimum 12.5 Gb bandwidth per GPU).

Management

- SN2201 switch per SU (8 switches total).
- Each SN2201 uplinked to core via 2x 100G links.
- 200 GbE management fabric with HA using sixteen 100 GbE core connections.
- All nodes connected to 1 Gb management switches.

Additional Considerations

- VLAN isolation separates fabrics on Converged (North/South) Network.
- Rack layout must provide power supply redundancy.
- HGX B300 servers per rack depends on available rack power.
- Dual-ported optic recommended to simplify breakout at NVIDIA ConnectX-8 OSFP port.

128 Nodes with 1024 B300 HGX GPUs

7. Software

Infrastructure software is a critical component of the NVIDIA Enterprise Reference Architecture, providing the foundation for deploying, managing, and orchestrating AI workloads at scale. In conjunction with the NVIDIA HGX AI Factory, the NVIDIA Enterprise AI Factory Validated Design (for enterprise and government) utilizes software tools and solutions from NVIDIA to provide a comprehensive, high-performance solution for modern AI projects and ensure optimal performance while adhering to best practices. Additionally, it includes tested and validated solutions from an ecosystem of partners, with specified versions and integrations, to support the end-to-end AI pipeline, from data curation to model testing, workflow orchestration, guard railing, security, and more.

Key components of these software designs include:

- **NVIDIA AI Enterprise**, a suite of software tools, libraries and frameworks, including NVIDIA NIM and NeMo microservices, that accelerate and simplify the development, deployment, and scaling of AI applications. Also includes **Base Command Manager**, which streamlines infrastructure provisioning, workload management and resource monitoring.
 - **NVIDIA Run:ai**, an orchestration platform that accelerates AI operations with dynamic orchestration across the AI lifecycle.
 - **NVIDIA NetQ**, a highly scalable, modern network operations toolset that provides visibility, troubleshooting and validation of Cumulus fabrics in real time.
-

8. NVIDIA-Certified Storage

This Enterprise Reference Architecture is designed to support high-performance storage solutions that can keep pace with GPU-intensive AI workloads. As a guideline, the architecture allocates approximately 12.5 Gb/s per GPU, scaling linearly as the cluster grows — for example, a 16-GPU cluster would require around 200 Gb/s of aggregate storage bandwidth. For validated storage systems and deployment guidance tailored to your cluster size, refer to the NVIDIA-Certified Storage program, which integrates seamlessly with this reference architecture.

The NVIDIA-Certified Storage program provides a comprehensive validation framework that ensures storage platforms meet stringent performance, quality, and interoperability standards for AI workloads. Partner solutions undergo extensive functional and performance testing to guarantee predictable deployment, energy efficiency, and operational consistency for end-to-end AI factories.

9. Services

A successful deployment requires more than advanced infrastructure — it also requires comprehensive services that accelerate deployment, ensure reliability, and achieve optimal performance.

NVIDIA works closely with its global partners to extend service reach, scale delivery, and provide deep industry expertise across all stages of implementation and operation.

- **Roadmap services** — Conduct current infrastructure assessments, model future capacity scenarios and develop phased plans to guide design, build, migration and operations of data centers over time.
- **First project implementation services** — Execute the initial deployment (from installation, configuration and commissioning to operational hand-over) of a first data center build or module.
- **Managed services** — Operate the ongoing monitoring, maintenance, and optimization of the data center including 24/7 support, contract management and vendor coordination.
- **Support Services** — Minimize downtime by providing expert guidance and access to NVIDIA specialists across compute, software and networking, ensuring optimal performance and availability.
- **Education Services** — Deliver online and hands-on training to customers and partners, providing in-depth knowledge that enables customers to maximize the value of their AI investments and partners to build deep AI expertise.

For additional information about NVIDIA Enterprise Services, visit <https://www.nvidia.com/en-us/support/enterprise/services/>.

10. Summary

The NVIDIA Enterprise Reference Architecture is the leading enterprise-scale architecture to meet the demanding and growing needs of AI compute. The Enterprise RA represents the architecture used by NVIDIA for internal AI model training, inference for reasoning AI, and large-scale data analytics.

This specific Enterprise RA focuses on the NVIDIA B300 baseboard delivered in HGX servers. It is interconnected using NVIDIA Spectrum-X for compute and converged connectivity. It includes guidance for both dual-plane and cost-effective single-plane network topologies to suit a variety of deployment needs.

The Enterprise RA documents a complete system of hardware and infrastructure software necessary to implement data centers for high-performance AI functionality. It leverages decades of NVIDIA expertise in accelerated computing, networking and AI software to remove guesswork, improve speed-to-value, optimize performance and TCO, and ensure enterprise-grade reliability and security.

The Enterprise RA is designed to both support the workloads of today and grow to support tomorrow's applications.

11. Appendix: Node Configurations

This appendix provides information on the compute and control plane node configurations.

Compute Nodes

The starting point and the basic building block of this Enterprise RA is the NVIDIA-Certified HGX B300 platform. This is covered in more detail in the Node Hardware section of this document but for completeness in the Appendix there is a more simplified version below.

Table 8: HGX B300 Server system components

Component	Quantity	Description
GPUs	8 on baseboard	Eight NVIDIA B300 SXM GPUs on an NVIDIA B300 baseboard, with GPU Memory of up to 2304 GB
DPU	1	NVIDIA BlueField-3 B3240 DPU with 2x 400G ports and 1 Gb RJ45 management port
SuperNIC	8 on baseboard	NVIDIA ConnectX-8 SuperNIC with a dual-port 400G configuration
CPUs	2	Minimum of 32 physical CPU cores per socket. Recommendation of 56 physical CPU cores per socket
System Memory	1	Minimum of 2 TB system memory
Storage	3	Inference Servers: minimum 1 TB NVMe drive per CPU socket. Training / DL Servers: minimum 2 TB NVMe drive per CPU socket. HPC Servers: minimum 1 TB NVMe drive per CPU socket. 1x1 TB NVMe boot drive
BMC/iLO	1	1 GbE RJ45 management port

Switches and Cables

Table 9 and Table 10 provide an overview of the switches and cables used in the cluster for a **dual-plane configuration**. The network adapters of the NVIDIA HGX B300 server, the storage servers, and the control plane servers are included for better understanding of the connectivity requirements.

Table 9: Switching with required inter-switch links, transceivers, and cables

Component	32 Server Nodes	64 Server Nodes	128 Server Nodes
NVIDIA HGX B300 SXM GPUs (8/node, on baseboard)	256	512	1024
NVIDIA ConnectX-8 SuperNIC, Compute fabric (8/node, on baseboard)	256	512	1024
BlueField-3 B3240 DPU, In-band management, Customer & Storage fabric and Support Servers	32	64	128
NVIDIA Spectrum-4 SN5600 Ethernet switch, compute core fabric	12	24	48
NVIDIA Spectrum-4 SN5600 Ethernet switch, converged core fabric	12	24	48
SN2201 leaf switches for OOB management fabric	4	8	16
OSFP, 2x 400G transceiver used for N/S inter-switch links (ISL)	26	256	512
Cable used for N/S inter-switch links (ISL)	26	256	512
OSFP, 2x 400G transceiver used for E/W inter-switch links (E/W ISL)	512	1024	2048
Cable used for E/W inter-switch links (ISL)	512	1024	2048
QSFP 100G transceivers used for OOB leaf switches	8	16	32
OSFP, 2x 400G transceiver used for OOB leaf switches	2	4	8
Cable used for OOB leaf switches	2	4	8

Table 10: End-point connections with required transceivers, and cables (dual plane)

Component	32 Server Nodes	64 Server Nodes	128 Server Nodes
NVIDIA HGX B300 SXM GPUs (8/node, on baseboard)	256	512	1024
NVIDIA ConnectX-8 SuperNIC, Compute fabric (8/node, on baseboard)	256	512	1024
BlueField-3 B3240 DPU, In-band management, Customer & Storage fabric and Support Servers	32	64	128
OSFP, 2x 400G transceiver used for Switch to compute node (DPU)	32	64	128
OSFP, 2x 400G transceiver used for Switch to compute node (SuperNIC)	256	512	1024
OSFP, 2x 400G transceiver used for SuperNIC to switch	256	512	1024
QSFP, 400G transceiver used for DPU to switch	64	128	256
Cable for Switch to ConnectX-8 SuperNIC	512	1024	2048
Cable for Switch to B3220 DPU	64	128	256
OSFP, 2x 400G transceiver used for switch to Storage	4	8	16
QSFP 100G transceiver used for upstream storage	32	64	128
Cable for storage	8	16	32

OSFP, 2x 400G transceiver used for switch to customer network	8	16	32
QSFP 100G transceiver used for customer network	64	128	256
Cable for customer network	16	32	64
OSFP, 2x 400G transceiver used for switch to control-plane B3220 DPUs	4	4	4
QSFP, 400G transceiver used for control plane B3220 DPUs	16	16	16
Cable for control-plane	8	8	8
CAT6 RJ45 cable for 1G OOB fabric	192	384	768

12. Notices

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